

2-D Navigation Dynamics, Linearization, and Multi-Rate Kalman Observer for INS/GPS Sensor Fusion

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December 8, 2025

1 3-D Nonlinear INS Model

We recall the standard nonlinear INS model given in equation (9) of Nemra & Aouf (IEEE Sensors Journal, 2010), expressed in compact form as

$$\dot{\mathbf{x}}^{\text{INS}}(t) = f(\mathbf{x}^{\text{INS}}(t), \mathbf{u}(t)), \quad (1)$$

where the state and input are

$$\mathbf{x}^{\text{INS}} = [\mathbf{p}_n^\top \quad \mathbf{v}_n^\top \quad \boldsymbol{\eta}^\top]^\top \in \mathbb{R}^9, \quad \mathbf{u} = [\mathbf{f}_b^\top \quad \boldsymbol{\omega}_b^\top]^\top \in \mathbb{R}^6. \quad (2)$$

Here

- $\mathbf{p}_n = [x_N, y_E, z_D]^\top$ is position in the navigation frame (e.g. NED: North–East–Down),
- $\mathbf{v}_n = [v_N, v_E, v_D]^\top$ is velocity in the navigation frame,
- $\boldsymbol{\eta} = [\phi, \theta, \psi]^\top$ is the Euler angle vector (roll, pitch, yaw),
- $\mathbf{f}_b = [a_x^b, a_y^b, a_z^b]^\top$ is the specific force measured by the IMU in the body frame,
- $\boldsymbol{\omega}_b = [p, q, r]^\top$ is the angular rate measured by the gyros in the body frame.

The standard INS mechanization equations can be written as

$$\dot{\mathbf{p}}_n = \mathbf{v}_n, \quad (3a)$$

$$\dot{\mathbf{v}}_n = C_b^n(\phi, \theta, \psi) \mathbf{f}_b + \mathbf{g}_n, \quad (3b)$$

$$\dot{\boldsymbol{\eta}} = T(\boldsymbol{\eta}) \boldsymbol{\omega}_b, \quad (3c)$$

where

- $C_b^n(\phi, \theta, \psi) \in \mathbb{R}^{3 \times 3}$ is the direction cosine matrix (DCM) from body to navigation frame (given in eq. (8) in the paper),
- $\mathbf{g}_n$ is the gravity vector expressed in the navigation frame (approximately $\mathbf{g}_n = [0, 0, g]^\top$ in NED with $g \approx 9.81$),
- $T(\boldsymbol{\eta}) \in \mathbb{R}^{3 \times 3}$ is the kinematic transformation mapping body angular rates to Euler angle rates (given by eq. (4) in the paper).

The observation (measurement) model corresponding to the INS-based position is given in equation (10) of the paper and has the generic form

$$\mathbf{y}_{\text{INS}}(t) = C_{\text{INS}} \mathbf{x}^{\text{INS}}(t), \quad (4)$$

where C_{INS} extracts the position components from $\mathbf{x}^{\text{INS}}$.

2 Derivation of a 2-D Nonlinear Navigation Model from the 3-D INS

We now derive a planar (2-D) navigation model from (1) under standard flat-earth and small-attitude assumptions.

2.1 Planar Assumptions

We assume:

1. **Flat earth and constant altitude.** The Down position z_D and vertical velocity v_D are approximately constant:

$$z_D(t) \approx z_{D0}, \quad v_D(t) \approx 0. \quad (5)$$

2. **Small roll and pitch angles.** The aircraft executes approximately coordinated, wings-level flight so that

$$\phi(t) \approx 0, \quad \theta(t) \approx 0. \quad (6)$$

3. **Yaw (ψ) is the only attitude variable relevant to planar navigation.** The heading dynamics are driven by the yaw rate r :

$$\dot{\psi}(t) = r(t). \quad (7)$$

4. **Gravity compensation in the IMU/INS.** The vertical channel (altitude control) compensates gravity such that the horizontal components of $C_b^n \mathbf{f}_b$ represent the net horizontal acceleration.

Under assumptions (2) and (3), the DCM $C_b^n(\phi, \theta, \psi)$ reduces in the horizontal plane to a pure yaw rotation:

$$C_b^n(\psi) \approx \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (8)$$

Let the horizontal position and velocity in the navigation frame be

$$\mathbf{p} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_N \\ y_E \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} v_N \\ v_E \end{bmatrix}. \quad (9)$$

Let the horizontal components of the body-frame specific force be

$$\mathbf{f}_b^{2D} = \begin{bmatrix} a_x^b \\ a_y^b \end{bmatrix}, \quad (10)$$

and let r denote the yaw rate.

Using (3b) together with (8), the horizontal acceleration components in the navigation frame are

$$\dot{v}_x = a_x^b \cos \psi - a_y^b \sin \psi, \quad (11a)$$

$$\dot{v}_y = a_x^b \sin \psi + a_y^b \cos \psi. \quad (11b)$$

From (3a), the position dynamics are

$$\dot{x} = v_x, \quad \dot{y} = v_y. \quad (12)$$

The yaw dynamics are simply

$$\dot{\psi} = r. \quad (13)$$

2.2 2-D Nonlinear Navigation State-Space Model

Define the 2-D navigation state vector

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ v_x \\ v_y \\ \psi \end{bmatrix} \in \mathbb{R}^5, \quad (14)$$

and the IMU input vector

$$\mathbf{u} = \begin{bmatrix} a_x^b \\ a_y^b \\ r \end{bmatrix} \in \mathbb{R}^3. \quad (15)$$

Combining (12), (11a), (11b), and (13) yields the 2-D nonlinear navigation dynamics

$$\dot{x} = v_x, \quad (16a)$$

$$\dot{y} = v_y, \quad (16b)$$

$$\dot{v}_x = a_x^b \cos \psi - a_y^b \sin \psi, \quad (16c)$$

$$\dot{v}_y = a_x^b \sin \psi + a_y^b \cos \psi, \quad (16d)$$

$$\dot{\psi} = r. \quad (16e)$$

In vector form,

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)), \quad (17)$$

where $f : \mathbb{R}^5 \times \mathbb{R}^3 \rightarrow \mathbb{R}^5$ is defined by the right-hand sides of (16a)–(16e).

2.3 GPS Measurement Model

Assume that the GPS provides direct measurements of the horizontal position in the navigation frame at a sampling period $T_{\text{GPS}} = 0.1$ s (10 Hz). The measurement model is

$$\mathbf{y}_{\text{GPS}}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} + \mathbf{v}_{\text{GPS}}(t), \quad (18)$$

or in linear form,

$$\mathbf{y}_{\text{GPS}}(t) = C \mathbf{x}(t) + \mathbf{v}_{\text{GPS}}(t), \quad (19)$$

with

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{v}_{\text{GPS}}(t) \sim \mathcal{N}(\mathbf{0}, R), \quad (20)$$

where $R \in \mathbb{R}^{2 \times 2}$ is the GPS measurement noise covariance.

3 Linearization of the 2-D Nonlinear Navigation Dynamics

We linearize (17) about a nominal trajectory $(\mathbf{x}_0(t), \mathbf{u}_0(t))$, which may represent, for example, straight and level flight at constant speed.

3.1 Perturbation Variables

Define the perturbation (or error) variables

$$\delta \mathbf{x}(t) := \mathbf{x}(t) - \mathbf{x}_0(t), \quad \delta \mathbf{u}(t) := \mathbf{u}(t) - \mathbf{u}_0(t). \quad (21)$$

The linearized dynamics have the form

$$\delta \dot{\mathbf{x}}(t) = A(t) \delta \mathbf{x}(t) + B(t) \delta \mathbf{u}(t), \quad (22)$$

where

$$A(t) = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}_0(t), \mathbf{u}_0(t)}, \quad B(t) = \left. \frac{\partial f}{\partial \mathbf{u}} \right|_{\mathbf{x}_0(t), \mathbf{u}_0(t)}. \quad (23)$$

Write the nominal state and input explicitly as

$$\mathbf{x}_0 = \begin{bmatrix} x_0 \\ y_0 \\ v_{x0} \\ v_{y0} \\ \psi_0 \end{bmatrix}, \quad \mathbf{u}_0 = \begin{bmatrix} a_{x0}^b \\ a_{y0}^b \\ r_0 \end{bmatrix}. \quad (24)$$

3.2 Jacobian Matrix $A(t)$

From (16a)–(16e),

$$\begin{aligned} f_1(\mathbf{x}, \mathbf{u}) &= v_x, \\ f_2(\mathbf{x}, \mathbf{u}) &= v_y, \\ f_3(\mathbf{x}, \mathbf{u}) &= a_x^b \cos \psi - a_y^b \sin \psi, \\ f_4(\mathbf{x}, \mathbf{u}) &= a_x^b \sin \psi + a_y^b \cos \psi, \\ f_5(\mathbf{x}, \mathbf{u}) &= r. \end{aligned}$$

The partial derivatives with respect to $\mathbf{x}$ are

$$\begin{aligned} \frac{\partial f_1}{\partial \mathbf{x}} &= [0 \quad 0 \quad 1 \quad 0 \quad 0], \\ \frac{\partial f_2}{\partial \mathbf{x}} &= [0 \quad 0 \quad 0 \quad 1 \quad 0], \\ \frac{\partial f_3}{\partial \mathbf{x}} &= [0 \quad 0 \quad 0 \quad 0 \quad -a_x^b \sin \psi - a_y^b \cos \psi], \\ \frac{\partial f_4}{\partial \mathbf{x}} &= [0 \quad 0 \quad 0 \quad 0 \quad a_x^b \cos \psi - a_y^b \sin \psi], \\ \frac{\partial f_5}{\partial \mathbf{x}} &= [0 \quad 0 \quad 0 \quad 0 \quad 0]. \end{aligned}$$

Evaluating at the nominal trajectory $(\mathbf{x}_0, \mathbf{u}_0)$, we obtain

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -a_{x0}^b \sin \psi_0 - a_{y0}^b \cos \psi_0 \\ 0 & 0 & 0 & 0 & a_{x0}^b \cos \psi_0 - a_{y0}^b \sin \psi_0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (25)$$

3.3 Jacobian Matrix $B(t)$

The partial derivatives with respect to $\mathbf{u}$ are

$$\begin{aligned} \frac{\partial f_1}{\partial \mathbf{u}} &= [0 \quad 0 \quad 0], \\ \frac{\partial f_2}{\partial \mathbf{u}} &= [0 \quad 0 \quad 0], \\ \frac{\partial f_3}{\partial \mathbf{u}} &= [\cos \psi \quad -\sin \psi \quad 0], \\ \frac{\partial f_4}{\partial \mathbf{u}} &= [\sin \psi \quad \cos \psi \quad 0], \\ \frac{\partial f_5}{\partial \mathbf{u}} &= [0 \quad 0 \quad 1]. \end{aligned}$$

Evaluated at $(\mathbf{x}_0, \mathbf{u}_0)$ we obtain

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \cos \psi_0 & -\sin \psi_0 & 0 \\ \sin \psi_0 & \cos \psi_0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (26)$$

3.4 Linearized Output Equation

The GPS output (19) is already linear in $\mathbf{x}$, so the linearized measurement model for perturbations is

$$\delta \mathbf{y}(t) = C \delta \mathbf{x}(t) + \mathbf{v}_{\text{GPS}}(t), \quad (27)$$

with the same matrix C as in (19).

4 Discretization and Multi-Rate Model (100 Hz IMU, 10 Hz GPS)

We discretize the linearized continuous-time model (22)–(27) with respect to the IMU sampling interval and then incorporate the slower GPS sampling.

4.1 IMU (Process) Sampling at 100 Hz

Let the IMU sampling period be

$$T_{\text{IMU}} = 0.01 \text{ s} \quad (100 \text{ Hz}), \quad (28)$$

and define discrete time instants

$$t_k = k T_{\text{IMU}}, \quad k \in \mathbb{Z}_{\geq 0}. \quad (29)$$

Assuming A and B are slowly varying or frozen over each sampling interval, the linearized dynamics discretize to

$$\delta \mathbf{x}_{k+1} = A_d \delta \mathbf{x}_k + B_d \delta \mathbf{u}_k + \mathbf{w}_k, \quad (30)$$

where

$$A_d = e^{A T_{\text{IMU}}}, \quad (31)$$

$$B_d = \int_0^{T_{\text{IMU}}} e^{A\tau} B d\tau, \quad (32)$$

and $\mathbf{w}_k$ represents process noise (including unmodelled dynamics and IMU noise), typically modelled as

$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, Q), \quad (33)$$

with $Q \in \mathbb{R}^{5 \times 5}$.

For small T_{IMU} , one often uses the first-order approximations

$$A_d \approx I + A T_{\text{IMU}}, \quad B_d \approx B T_{\text{IMU}}, \quad (34)$$

and absorbs the approximation error into Q .

4.2 GPS (Measurement) Sampling at 10 Hz

Let the GPS sampling period be

$$T_{\text{GPS}} = 0.1 \text{ s} = 10 T_{\text{IMU}}, \quad (35)$$

and define GPS measurement instants

$$t_\ell^{\text{GPS}} = \ell T_{\text{GPS}} = (10\ell) T_{\text{IMU}}, \quad \ell \in \mathbb{Z}_{\geq 0}. \quad (36)$$

Thus each GPS measurement time t_ℓ^{GPS} coincides with IMU time index $k_\ell = 10\ell$.

The discrete-time measurement equation at GPS instants is

$$\delta \mathbf{y}_\ell = C \delta \mathbf{x}_{k_\ell} + \mathbf{v}_\ell, \quad (37)$$

where

$$\mathbf{v}_\ell \sim \mathcal{N}(\mathbf{0}, R). \quad (38)$$

For IMU time indices k that are not multiples of ten (i.e. $k \neq k_\ell$ for all ℓ), no GPS measurement is available, so only the process model (30) is used.

5 Closed-Loop Observer and Kalman Filter Design

We now design a discrete-time observer (Luenberger/Kalman) for the 2-D navigation system using the multi-rate sensor configuration.

5.1 Luenberger-Type Closed-Loop Observer (Generic Gain)

Consider the discrete-time linear system at IMU rate:

$$\mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d \mathbf{u}_k + \mathbf{w}_k, \quad (39a)$$

$$\mathbf{y}_k = C_k \mathbf{x}_k + \mathbf{v}_k. \quad (39b)$$

Here, to capture the multi-rate nature,

$$C_k = \begin{cases} C, & \text{if a GPS measurement is available at time } k, \\ 0, & \text{otherwise,} \end{cases} \quad (40)$$

and $\mathbf{y}_k$ is taken to be the two-dimensional GPS measurement vector when available and $\mathbf{y}_k = \mathbf{0}$ when not.

A general closed-loop observer with time-varying gain L_k is

$$\hat{\mathbf{x}}_{k+1} = A_d \hat{\mathbf{x}}_k + B_d \mathbf{u}_k + L_k (\mathbf{y}_k - C_k \hat{\mathbf{x}}_k), \quad (41)$$

where $\hat{\mathbf{x}}_k$ is the state estimate at time k .

Define the estimation error $\mathbf{e}_k = \mathbf{x}_k - \hat{\mathbf{x}}_k$. Using (39a)–(39b) and (41), the error dynamics are

$$\mathbf{e}_{k+1} = (A_d - L_k C_k) \mathbf{e}_k + \mathbf{w}_k - L_k \mathbf{v}_k. \quad (42)$$

If L_k is chosen such that the eigenvalues of $A_d - L_k C_k$ lie strictly inside the unit circle whenever $C_k = C$, the observer is stable in the deterministic sense. The Kalman filter chooses L_k to minimize the estimation error covariance, as described next.

5.2 Discrete-Time Kalman Filter with Multi-Rate Measurements

Assume $\mathbf{w}_k$ and $\mathbf{v}_k$ are zero-mean, white, mutually independent Gaussian noise processes with covariances

$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, Q), \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, R_k), \quad (43)$$

where

$$R_k = \begin{cases} R, & \text{if a GPS measurement is present at time } k, \\ \text{undefined,} & \text{otherwise (no measurement).} \end{cases} \quad (44)$$

The Kalman filter is naturally expressed in predictor–corrector form. It provides both the state estimate and the error covariance estimate.

Prediction (Time Update): Every IMU Step (100 Hz)

For each IMU time step k :

$$\hat{\mathbf{x}}_{k+1|k} = A_d \hat{\mathbf{x}}_{k|k} + B_d \mathbf{u}_k, \quad (45a)$$

$$P_{k+1|k} = A_d P_{k|k} A_d^\top + Q, \quad (45b)$$

where

- $\hat{\mathbf{x}}_{k|k}$ is the state estimate at time t_k given all available measurements up to time t_k ,
- $P_{k|k}$ is the estimation error covariance at time t_k ,
- $\hat{\mathbf{x}}_{k+1|k}$ and $P_{k+1|k}$ are the one-step predicted state and covariance using the process model.

Correction (Measurement Update): Only When GPS is Available (10 Hz)

Let $\mathcal{K}_{\text{GPS}} \subset \mathbb{Z}_{\geq 0}$ be the set of IMU indices at which a GPS measurement is available (i.e. $k \in \mathcal{K}_{\text{GPS}}$ if $k = 10\ell$ for some ℓ).

For $k \in \mathcal{K}_{\text{GPS}}$ (GPS update step), with measurement $\mathbf{y}_k$ obeying

$$\mathbf{y}_k = C \mathbf{x}_k + \mathbf{v}_k, \quad (46)$$

the Kalman correction step is

$$K_k = P_{k|k-1} C^\top (C P_{k|k-1} C^\top + R)^{-1}, \quad (47a)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + K_k (\mathbf{y}_k - C \hat{\mathbf{x}}_{k|k-1}), \quad (47b)$$

$$P_{k|k} = (I - K_k C) P_{k|k-1}, \quad (47c)$$

where K_k is the Kalman gain.

For $k \notin \mathcal{K}_{\text{GPS}}$ (no GPS measurement), the predicted values become the filtered values:

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1}, \quad P_{k|k} = P_{k|k-1}. \quad (48)$$

Closed-Loop Observer Interpretation

The Kalman filter recursion (45a)–(47c) can be written in the Luenberger observer form (41) by identifying the Kalman gain with the observer gain:

$$L_k = \begin{cases} K_k, & k \in \mathcal{K}_{\text{GPS}}, \\ 0, & k \notin \mathcal{K}_{\text{GPS}}. \end{cases} \quad (49)$$

Thus, the Kalman filter is a particular closed-loop observer whose gain L_k is chosen optimally (in the least-squares sense) based on the noise covariances Q and R .

6 Summary

Starting from the 3-D INS model (equation (9) in Nemra & Aouf), we have

- derived a planar (2-D) nonlinear navigation model (16a)–(16e) for position, horizontal velocity, and yaw angle;
- computed the continuous-time linearization about a nominal trajectory, producing the Jacobian matrices A (25) and B (26);
- discretized the linearized model at the IMU sampling rate of 100 Hz (IMU as process input) and incorporated GPS measurements at 10 Hz as a multi-rate measurement process;
- designed a closed-loop observer in Luenberger form and specialized it to the discrete-time multi-rate Kalman filter (45a)–(47c).

The resulting algorithm combines IMU data (100 Hz) and GPS position (10 Hz) to provide an optimal 2-D navigation state estimate.